FLAT BASE CHANGE FOR THE COTANGENT COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the commutative square and the proof to the double square of the chapter; both are fixed by labelled equations and the displayed computation is inside the counted span.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The computation of the cotangent complex as a lower shriek of differentials. Lemma lemma-compute-cotangent-complex (545-555, produced as L44157), used at proof lines 1215-1220 and 1232-1233. Rewrites both sides as lower shrieks of differential sheaves.
- The change-of-rings lemma for lower shrieks. Cohomology on Sites, Lemma sites-cohomology-lemma-change-of-rings, used at proof lines 1221-1222. Turns the derived tensor product into a lower shriek along h.
- Flatness of the sheaf of differentials over the structure sheaf. used at proof lines 1223-1230. Computes the pullback of the differential sheaf as the differential sheaf of the primed square.
- Flat base change for the lower shriek in this setting. Lemma lemma-flat-base-change (1174-1181, produced as L44146), used at proof lines 1231-1233. Identifies the two lower shrieks and hence concludes the isomorphism.

Source reference. Original equation 08QM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11065–11065. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 08QN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11066–11066. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08PU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11037–11037. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11067–11067. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08Q8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11051–11051. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *If (08QM) induces a quasi-isomorphism $B \otimes^{\mathbf{L}}_A A' = B'$, then the functoriality map induces an isomorphism*

$$L_{B/A} \otimes^{\mathbf{L}}_B B' \longrightarrow L_{B'/A'}$$

Proof. We will use the notation introduced in Equation (08QN). We have

$$L_{B/A} \otimes^{\mathbf{L}}_B B' = L\pi_!(\Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B}) \otimes^{\mathbf{L}}_B B' = L\pi_!(Lh^*(\Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B}))$$

the first equality by Lemma 08PU and the second by Cohomology on Sites, Lemma 08Q8. Since $\Omega_{\mathcal{O}/A}$ is a flat $\mathcal{O}$-module, we see that $\Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B}$ is a flat $\underline{B}$-module. Thus $Lh^*(\Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B}) = \Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B'}$ which is equal to $g^{-1}(\Omega_{\mathcal{O}'/A'} \otimes_{\mathcal{O}'} \underline{B'})$ by inspection. we conclude by Lemma 08QP and the fact that $L_{B'/A'}$ is computed by $L\pi'_!(\Omega_{\mathcal{O}'/A'} \otimes_{\mathcal{O}'} \underline{B'})$. □